

Company: Infosys

Role: Systems Engineer

Round: SQL Practical + Logical Trap

Question 1

Create table `Employee`:

- `id INT`
- `name VARCHAR(50)`
- `salary DECIMAL(10,2)`
- `dept VARCHAR(30)`

Insert:

```
1, 'Ravi', 50000, 'IT'  
2, 'Ravi', 60000, 'HR'  
3, 'Anu', 45000, 'IT'
```

Now execute:

```
UPDATE Employee  
SET salary = 70000  
WHERE name = 'Ravi';
```

1. How many rows will be updated?
 2. Why?
 3. How would you update only one Ravi without using constraints?
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Question 2

You run:

```
DELETE FROM Employee;  
ROLLBACK;
```

Explain:

1. When will ROLLBACK work?
2. When will it fail?
3. What happens if auto-commit is enabled?

They are testing transaction awareness.

Company: Tata Consultancy Services

Role: Assistant System Engineer
Round: Scenario-Based SQL

Question 3

Create table `Orders`:

- `order_id` INT
- `customer` VARCHAR(50)
- `amount` DECIMAL(8,2)

Insert:

```
101, 'Kiran', 1000  
102, 'Arun', 2000  
103, 'Kiran', 3000
```

Now execute:

```
UPDATE Orders  
SET amount = amount + 500;
```

1. What is the result?
2. Why is this dangerous in production?
3. How should this query have been written?

If you forget WHERE, you're not production safe.

Question 4

Execute:

```
DELETE FROM Orders WHERE amount > 1500;
```

After deletion, how many rows remain?

Then execute:

```
ROLLBACK;
```

Will the deleted rows come back? Explain conditions clearly.

Company: Wipro

Role: Project Engineer
Round: Debugging SQL

Question 5

Create table `Students`:

- roll_no INT
- name VARCHAR(50)

- marks INT

Insert:

```
1, 'Rahul', 80
2, 'Priya', 90
3, 'Rahul', 60
```

Now run:

```
DELETE FROM Students
WHERE name = 'Rahul';
```

1. How many rows are deleted?
2. Why might this be a logical mistake?
3. What real-world issue can this cause?

This tests data uniqueness awareness without constraints.

Question 6

You run:

```
ALTER TABLE Students DROP marks;
```

Then immediately:

```
ROLLBACK;
```

Will the column return? Why or why not?

They want to see if you understand DDL auto-commit behavior.

Company: Accenture

Role: Associate Software Engineer

Round: Multi-Step Logic

Question 7

Create table `Payroll`:

- `emp_id` INT
- `emp_name` VARCHAR(50)
- `basic_salary` DECIMAL(10,2)
- `bonus` DECIMAL(10,2)

Insert 3 records.

Now execute in order:

```
UPDATE Payroll SET bonus = 5000 WHERE emp_id = 1;  
DELETE FROM Payroll WHERE emp_id = 2;  
ROLLBACK;
```

1. What is final table state?
2. Will both UPDATE and DELETE revert?
3. Under what condition?

Be precise. No vague answer.

Question 8

You rename table:

```
RENAME TABLE Payroll TO Employee_Pay;
```

Immediately after, you execute:

```
ROLLBACK;
```

Will table name revert? Explain clearly.

Most freshers answer this wrong.

Company: Amazon

Role: SDE Intern

Round: Edge Case Thinking

Question 9

Create table `Users`:

- `id INT`
- `username VARCHAR(50)`
- `salary DECIMAL(8,2)`

Insert:

```
1, 'alex', 50000
2, 'alex', 70000
```

Now execute:

```
UPDATE Users
SET salary = 60000
WHERE username = 'alex'
AND salary = 50000;
```

1. How many rows update?
2. Why is this safer than only using username condition?

Question 10

You execute:

```
DELETE FROM Users WHERE id = 10;
```

But id 10 does not exist.

1. Will error occur?
2. What will be affected rows count?